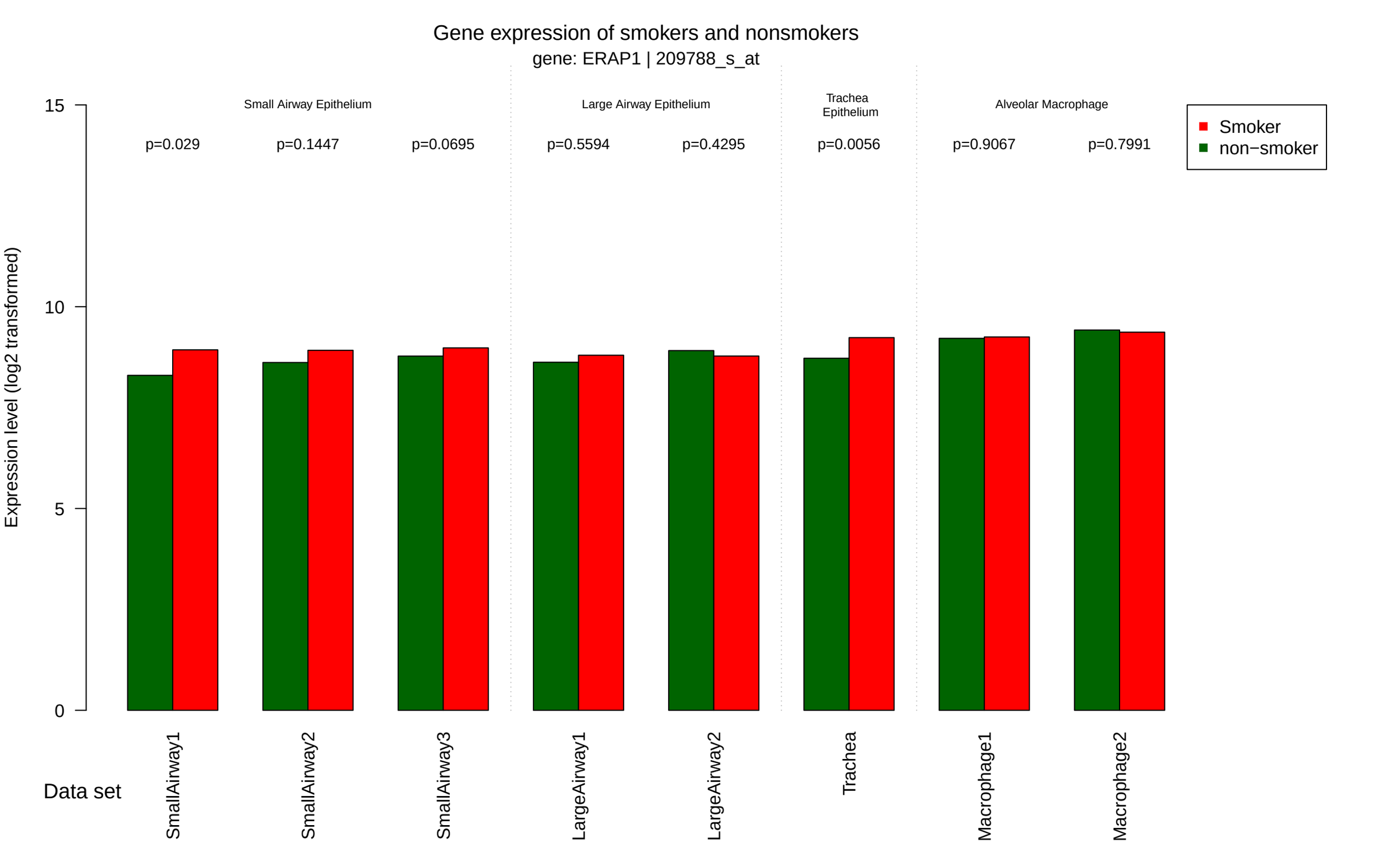
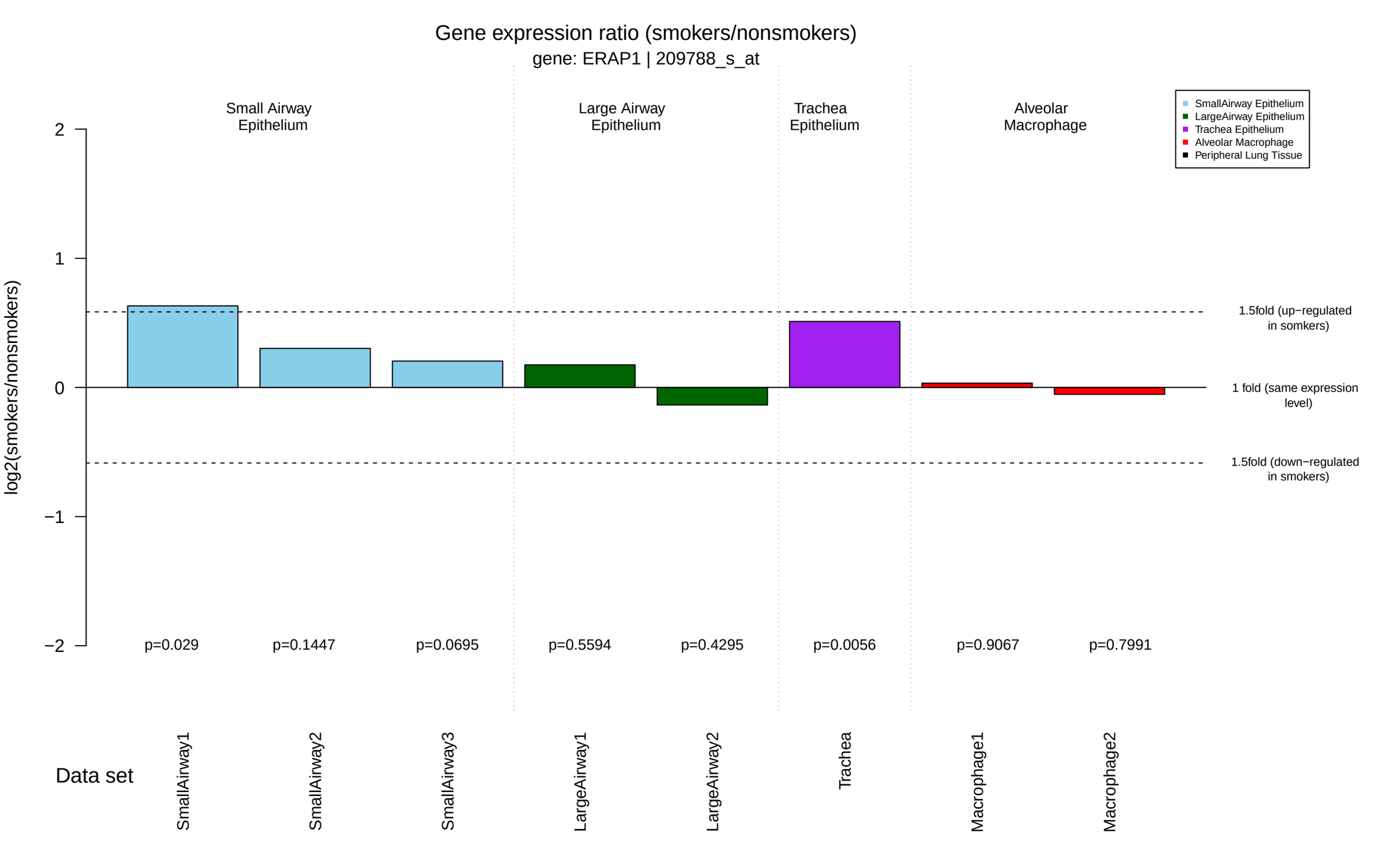
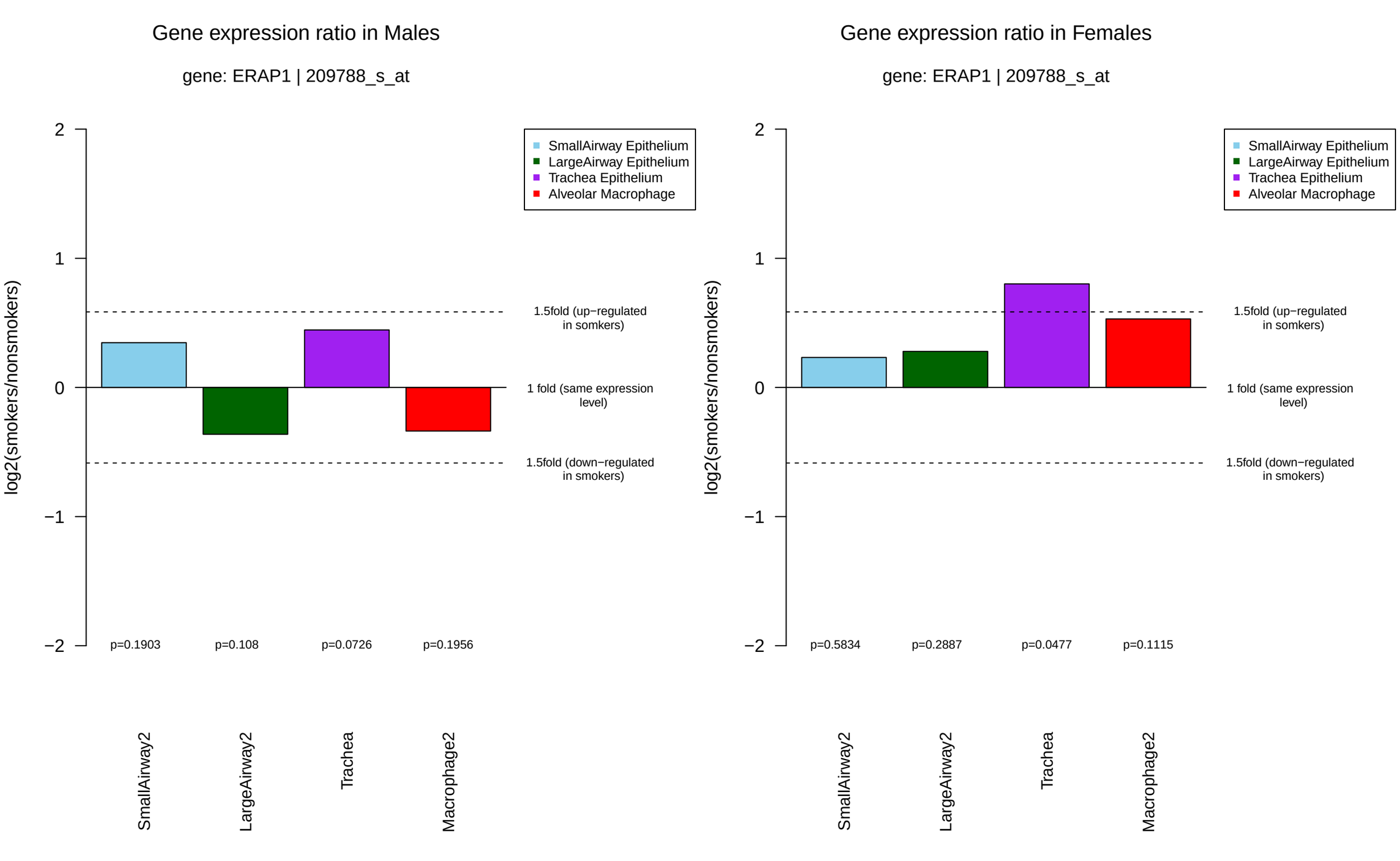
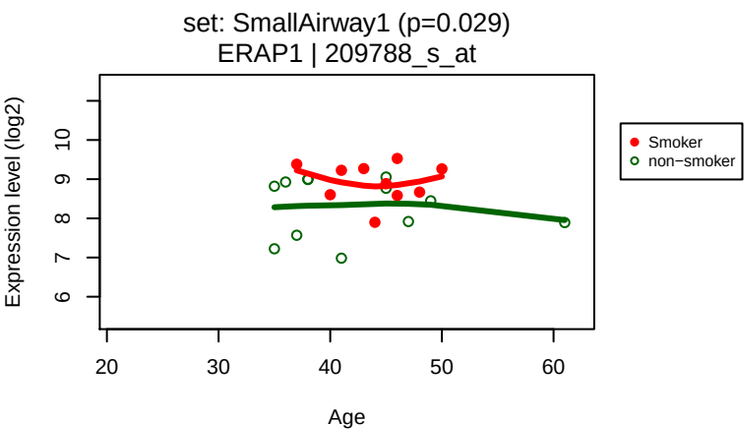
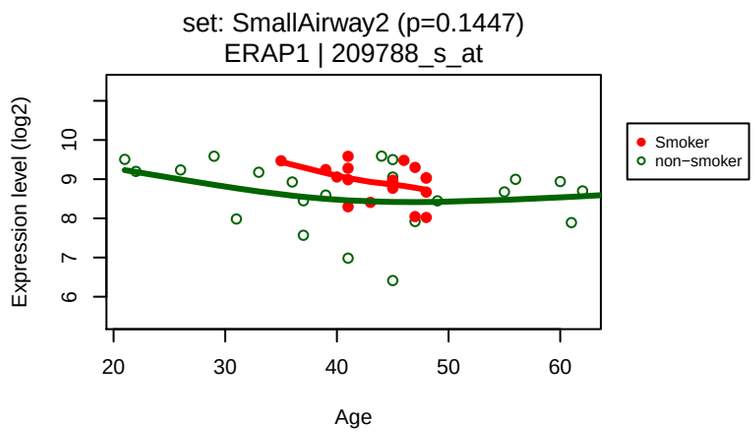
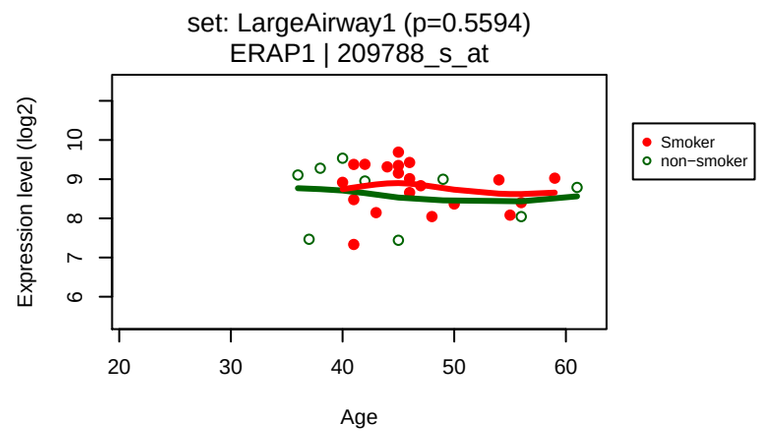

Gene expression of smokers and nonsmokers
gene: ERAP1 | 209788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.029
p=0.1447
p=0.0695
p=0.5594
p=0.4295
p=0.0056
p=0.9067
p=0.7991
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

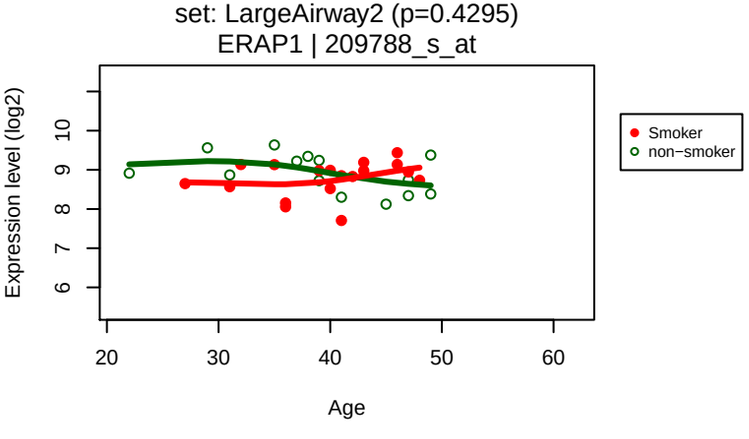

Gene expression ratio (smokers/nonsmokers)
gene: ERAP1 | 209788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.029
p=0.1447
p=0.0695
p=0.5594
p=0.4295
p=0.0056
p=0.9067
p=0.7991
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

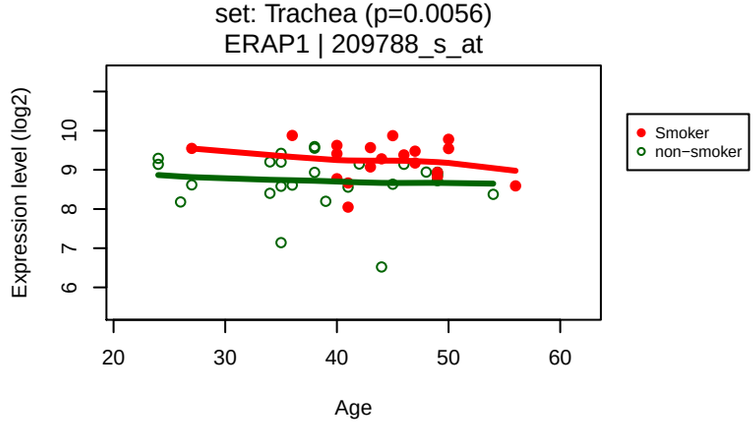

Gene expression ratio in Males
gene: ERAP1 | 209788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1903
p=0.108
p=0.0726
p=0.1956
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP1 | 209788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5834
p=0.2887
p=0.0477
p=0.1115
SmallAirway2
LargeAirway2
Trachea
Macrophage2

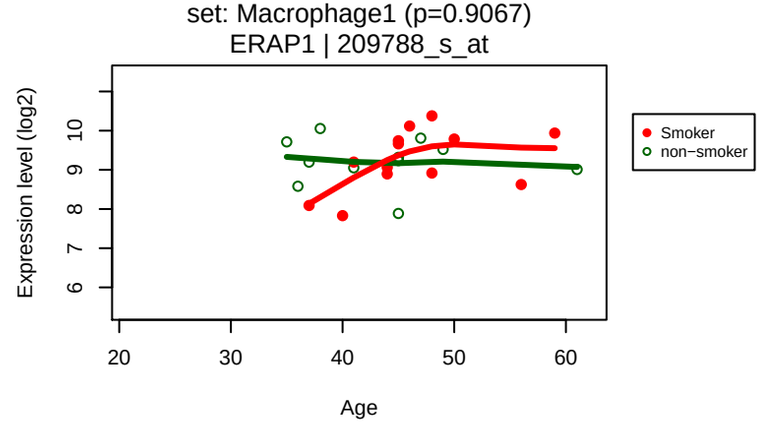

set: SmallAirway1 (p=0.029)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker

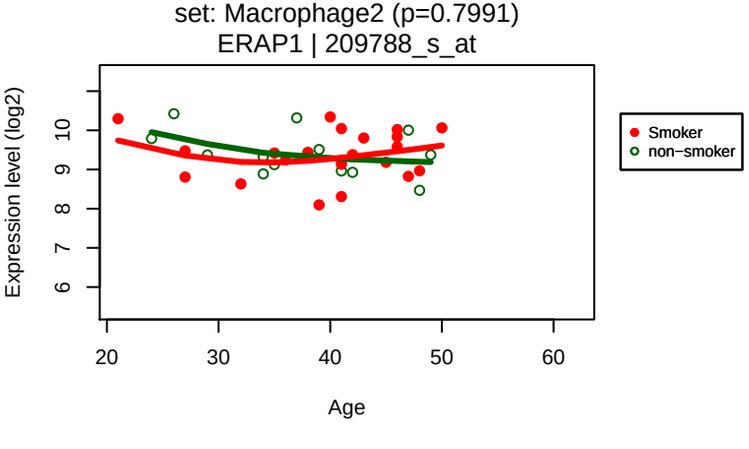

set: SmallAirway2 (p=0.1447)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker

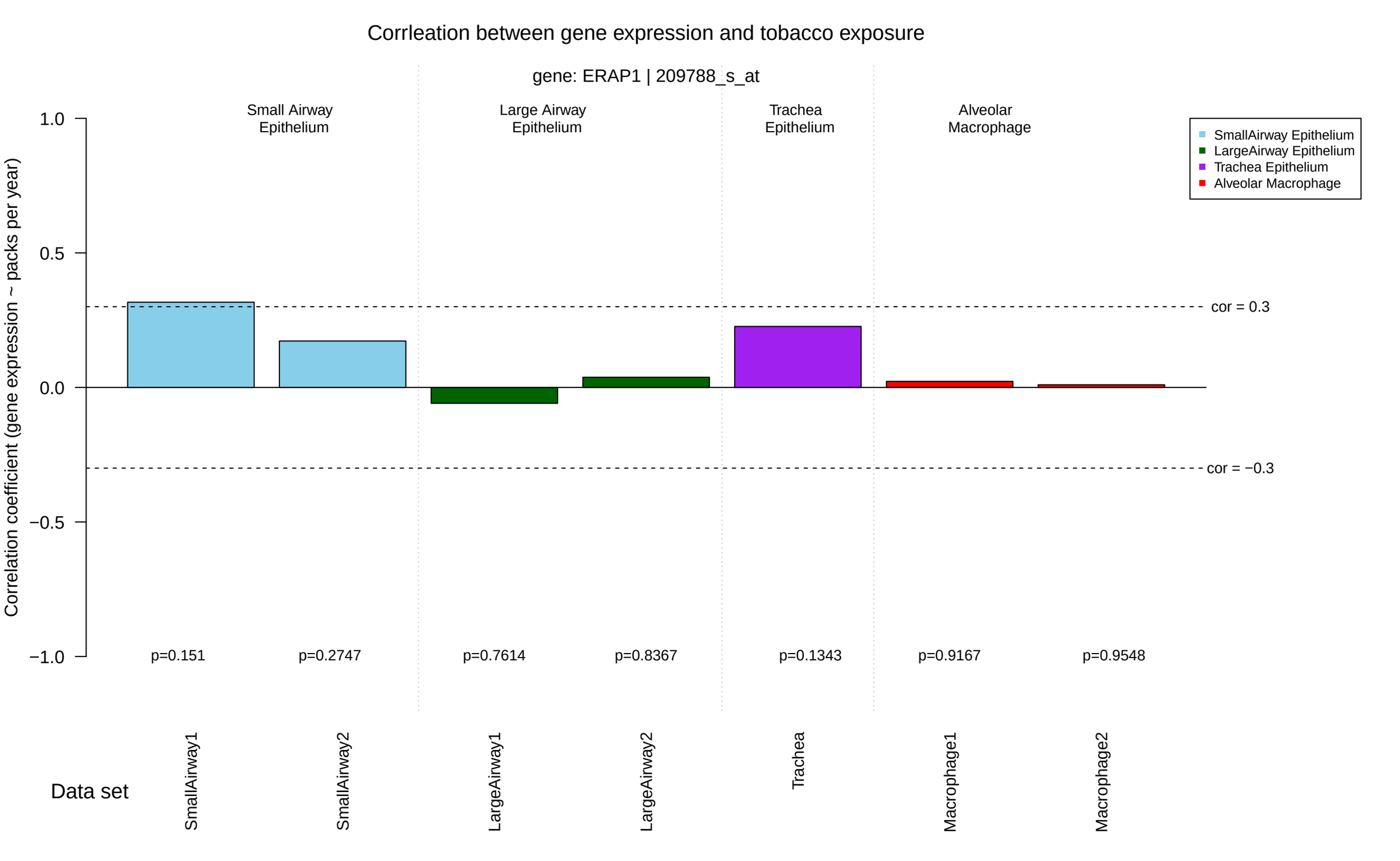

set: LargeAirway1 (p=0.5594)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4295)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0056)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9067)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7991)
ERAP1 | 209788_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP1 | 209788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.151
p=0.2747
p=0.7614
p=0.8367
p=0.1343
p=0.9167
p=0.9548
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2